

Daniel He

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Education

Western University - BSc., Computer Science

September 2024 - May 2028 Expected

With Admission Scholarship, Scholarship of Distinction (\$2500)

Deans Honour List 2024-2025 (**3.93/4.00 GPA**)

Work Experience

OpsGuru - Software Engineering Intern

Incoming May 2026-August 2026

AWS AI/ML Infrastructure & Agent Systems

Western University - Undergraduate Research Assistant

October 2025 - Present

LLM Hallucination - Dr. Apurva Narayan

- Co-authored *HalluciBench*, an LLM hallucination benchmarking paper, under review at COLM 2026
- Designed and implemented an LLM-as-a-judge benchmarking pipeline in Python with **Ollama** to evaluate hallucination rates across **29 LLMs**, incorporating Hyperparameter Tuning and **Prompt Engineering**
- Implemented a **Reinforcement Learning**-inspired **Synthetic Data** generation pipeline from a research paper to create a **1000-sample** dataset for evaluating LLM refusal calibration and hallucination rates

Scotiabank - Developer 1 (Intern)

May 2025 - August 2025

IT&S Capital Markets/Data Analytics

- Refactored a broken **Airflow** DAG pipeline to store vector embeddings in **ChromaDB** with persistence to **Google Cloud Storage**, migrating embedding logic into a Cloud Function. Cut p95 latency by ~90% and resolved a 6-month production issue to enable scalable AI workflows that support RAG for the bank's internal knowledge bot
- Enhanced internal research summarization pipeline by extending PDF parsing support from 2 to 4 document formats, automated deployment using **CI/CD** build configuration tools, and documented using Jira and Confluence
- Built a **Python** cyclomatic complexity analyzer using SonarQube and Selenium, reducing analysis time by ~85%

Projects

Dex2 ([GitHub Link](#), [Video-Demo](#))

February 2026

- Built a **TypeScript (React.js)** Chrome extension that auto-captures page content to trigger context-aware agent actions (edit slides, open tabs, draft emails) using **MongoDB** for embeddings and Supabase (**PostgreSQL**)
- Implemented AI **Agent Orchestration** using FastAPI for hybrid retrieval and **Tool Calling** with background workers and concurrency to keep chat non-blocking. Used **LangChain** for chunking and retrieval
- Built a Google Slides agent combining **OAuth**, GPT-4o-mini and Gemini vision to generate and edit slides directly in live presentations via the Slides API, using vision-based style extraction to match existing formatting

Mini-v ([GitHub Link](#))

April 2026

- Built a **C++** LLM inference server with micro-batching and continuous batching, achieving **2.5x throughput** improvement (6.5 to 16.8 req/s) across concurrent load with 100% request success rate

Say Less ([Devpost Link](#), [Live-Demo](#))

November 2025

Hack Trent 2025 - MLH Best Use of ElevenLabs (21 total submissions)

- Created a website that converts American Sign Language (ASL) to speech and speech to text in real time
- Spearheaded development of a **FastAPI** backend with **RESTful API** endpoints, integrating a fine tuned **MediaPipe** model (~80% accuracy on ASL to text). Containerized with **Docker** and deployed on **Render**

Skills

Languages: Python, Java, JavaScript, TypeScript, C, SQL

Frameworks and Libraries: Next.js, React.js, FastAPI, MediaPipe, Tensorflow, PyTorch, SQLAlchemy, LangChain

Developer Tools: Docker, Git, Render, Vercel, GCP, Apache Airflow, ChromaDB, SonarQube, Selenium, Ollama, Linux

Databases: PostgreSQL (Supabase), SQLite, MongoDB Atlas